

Chain Rule Worksheet

Twenty-four composition problems from basic powers through nested, product, and quotient combinations.

Revision: 2026-07-23 **Canonical page:** <https://bettergrades.net/subjects/math/calculus/worksheets/chain-rule/>

Name: _____ **Date:** _____

1. Differentiate $y = (3x + 2)^5$.

2. Differentiate $y = (5x - 1)^4$.

3. Differentiate $y = (2x + 4)^6$.

4. Differentiate $y = (-3x + 2)^3$.

5. Differentiate $y = (7x + 1)^2$.

6. Differentiate $y = (4x - 5)^3$.

7. Differentiate $y = e^{3x^2}$.

8. Differentiate $y = e^{\sin x}$.

9. Differentiate $y = \ln(5x - 2)$.

10. Differentiate $y = \ln(x^2 + 4)$.

11. Differentiate $y = \sin(4x^3)$.

12. Differentiate $y = \cos(2x - 1)$.

13. Differentiate $y = \tan(x^2)$.

14. Differentiate $y = \sin^3 x$.

15. Differentiate $y = (1 + (2x - 3)^2)^4$.

16. Differentiate $y = e^{\sqrt{1+x^2}}$.

17. Differentiate $y = \ln(1 + \sin^2 x)$.

18. Differentiate $y = x^2 e^{3x}$.

19. Differentiate $y = x \sin(x^2)$.

20. Differentiate $y = \frac{\ln x}{x^2}$.

21. Differentiate $y = \frac{e^x}{1+x^2}$.

22. Differentiate $y = (x^2 + 1)^3(x - 2)$.

23. Differentiate $y = \frac{\sin(2x)}{x}$.

24. Differentiate $y = \sqrt{1 + e^{2x}}$.